Carlos Manuel Carlevaro: *Curriculum Vitæ – short*

Personal information

Date of birth: August 31, 1968.
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Google Scholar profile
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Professional Profile

Experienced Physicist and Researcher specializing in Computational Physics, Granular Materials, Biophysics and Complex Systems. Extensive track record in developing numerical models for industrial and biological systems. Currently leading research initiatives as Director of Center for Experimental and Computational Mechanics Research and as an Independent Researcher at CONICET.

Current positions

2007 - Present **National Scientific and Technical Research Council** (CONICET).
Current position: *Independent Researcher.* manuel@iflysib.unlp.edu.ar
Institute of Physics of Liquids and Biological Systems (IFLYSIB). Calle 59 Nro. 789.
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2005 - Present **Universidad Tecnológica Nacional.**
Current position: *Full Professor – Head of Center for Experimental and Computational Mechanics Research.*
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La Plata Regional School, Mechanic Engineering Department. Avenida 60 esquina 124 s/n.
1923 Berisso, Buenos Aires, Argentina. Phone (+54 221) 421-4392.

Education

2002 **PhD in Exact Sciences** , National University of La Plata. Argentina.
Thesis: *Microscopic Model of Liquid Water. Generalized Mean Spherical Approximation.*
Advisor: Dr. Fernando Vericat.

1995 **Graduated in Physics**, National University of Rosario. Argentina.

Expertise & Research Interests

- **Granular Matter:** Silo discharge, clogging, stick-slip dynamics, transport.
- **Soft Matter & Fluids:** Pervaporation, structural and dynamics properties of confined fluids, nanochannel transport.
- **Biophysics:** CAR T-cell theraphy modeling, protein-lipid interactions.
- **Computational Modeling & Simulation:** DEM (Discrete Element Method), ABM (Agent Based Modeling), System Dynamics, Optimization (Genetic Algorithm and Swarm Particle Optimization).

Publications

Book

2025 F. Batista and Manuel Carlevaro. **Python para Ciencia y Tecnología**, 2025. Self-published, ISBN: 978-631-00-9794-7. https://libropython.science/.

Peer-reviewed Journal Publications (last three years – 55 total publications)

2026
13. R.G. Rossatto, H. Ariel Alvarez, C.M. Carlevaro, and José Rafael Bordin. "Confinement-controlled chase–escape dynamics". In: *Physica A: Statistical Mechanics and its Applications* 692 (2026), p. 131524. DOI: https://doi.org/10.1016/j.physa.2026.131524.
12. Mariano Martín Ramírez, Esteban Alejo Domené, and Manuel Carlevaro. "Transport of Fluid Mixtures Through Organic and Inorganic Nanochannels: The Main Influence of Organic and Inorganic Poral Throats on the Molecular Distributions and Hydrodynamics of Different Confined Fluid Mixtures". In: *Energy Technology* 14.1 (2026), e202501679. DOI: https://doi.org/10.1002/ente.202501679.

2025
11. Santiago Mosca, C. Manuel Carlevaro, and Enrique Lomba. "Molecular dynamics study of pervaporation of water–methanol mixtures through graphene nanotubes". In: *The Journal of Chemical Physics* 163.16 (Oct. 2025), p. 164501. DOI: 10.1063/5.0299861.
10. Yanis R. Espinosa, C. Manuel Carlevaro, and C. Gastón Ferrara. "Molecular mechanisms underlying the effects of urea and the structural dynamics of bovine serum albumin". In: *Biointerphases* 20.4 (July 2025), p. 041003. DOI: 10.1116/6.0004688.
9. Federico G. Vega, C. Manuel Carlevaro, Mauro Baldini, Marcos A. Madrid, and Luis A. Pugnaloni. "Simulation of proppant conductivity test: effect of particle size dispersion". In: *Petroleum Science and Technology* 43.9 (2025), pp. 1010–1028. DOI: 10.1080/10916466.2024.2326653.

2024
8. Amir Zablotsky, Marcos A. Madrid, C. Manuel Carlevaro, Marcelo Kuperman, Luis A. Pugnaloni, and Sebastián Bouzat. "Reduction of clogging of vibrated grains passing through a narrow aperture by the addition of low-friction particles". In: *Phys. Rev. E* 110 (3 Sept. 2024), p. 034902. DOI: 10.1103/PhysRevE.110.034902.
7. Luciana Melina Luque, Carlos Manuel Carlevaro, Enrique Rodriguez-Lomba, and Enrique Lomba. "In silico study of heterogeneous tumour-derived organoid response to CAR T-cell therapy". In: *Scientific Reports* 14.1 (May 29, 2024), p. 12307. DOI: 10.1038/s41598-024-63125-5.
6. C. Manuel Carlevaro, Ryan Kozlowski, and Luis A. Pugnaloni. "Flow rate in 2D silo discharge of binary granular mixtures: the role of ordering in monosized systems". In: *Frontiers in Soft Matter* 4 (2024). DOI: 10.3389/frsfm.2024.1340744.

2023
5. Rituparna Basak, Ryan Kozlowski, Luis A. Pugnaloni, M. Kramár, Joshua E. S. Socolar, C. Manuel Carlevaro, and Lou Kondic. "Evolution of force networks during stick-slip motion of an intruder in a granular material: Topological measures extracted from experimental data". In: *Phys. Rev. E* 108 (5 Nov. 2023), p. 054903. DOI: 10.1103/PhysRevE.108.054903.
4. María José Cervantes, Lucas O. Basiuk, Ana González-Suárez, C. Manuel Carlevaro, and Ramiro M. Irastorza. "Low-Frequency Electrical Conductivity of Trabecular Bone: Insights from In Silico Modeling". In: *Mathematics* 11.19 (2023). DOI: 10.3390/math11194038.
3. H. Ariel Alvarez, Alexandra Cousido-Siah, Yanis R. Espinosa, Alberto Podjarny, C. Manuel Carlevaro, and Eduardo Howard. "Lipid exchange in crystal-confined fatty acid binding proteins: X-ray evidence and molecular dynamics explanation". In: *Proteins: Structure, Function, and Bioinformatics* 91.11 (2023), pp. 1525–1534. DOI: https://doi.org/10.1002/prot.26546.
2. Marcia C. Barbosa, Ana Laura Benavides, Manuel Carlevaro, Gerhard Kahl, and Enrique Lomba. "Special issue on soft matter research in Latin America". In: *Journal of Physics: Condensed Matter* 35.41 (July 2023), p. 410301. DOI: 10.1088/1361-648X/acdebd.
1. Luciana Melina Luque, Carlos Manuel Carlevaro, Camilo Julio Llamoza Torres, and Enrique Lomba. "Physics-based tissue simulator to model multicellular systems: A study of liver regeneration and hepatocellular carcinoma recurrence". In: *PLOS Computational Biology* 19.3 (Mar. 2023), pp. 1–28. DOI: 10.1371/journal.pcbi.1010920.

Supervision and Mentoring

Postdoctoral Fellows

2023 – Martín Ramirez, Mariano Esteban. CONICET - YTEC.
2020 – 2023 Luque, Luciana Melina. CONICET.
2020 – 2022 Espinosa Silva, Yanis Ricardo. CONICET.

2017 – 2020 Lotto, Federico. CONICET.

2017 – 2019 Sánchez, Hernán Rubén. CONICET.

2017 – 2020 Vega, Federico. CONICET - YTEC.

PhD. Candidates

2024 – Tomás, Kevin Gabriel. «Control of non-linear dynamics in granular dissipative systems». National University of La Plata.

Gracia, César. «Impact of fluid viscosity on proppant transport and sedimentation in hydraulic stimulation of reservoirs». Universidad Tecnológica Nacional.

2023 – Montero, Julián. «Clogging of vibrated bidisperse granular media». National University of La Plata.

2020 – Mosca, Santiago. «Flow and transport modeling in porous media». Universidad Tecnológica Nacional.

Basiuk, Lucas Osvaldo. «Computational design of stochastically optimized scaffolds». Universidad Tecnológica Nacional.

Research Staff Supervision

2022 – Ferrara, Carlos Gastón.

2021 – Sánchez, Hernán Rubén. CONICET.

2019 – 2022 Madrid, Marcos Andrés. CONICET.

2015 – 2019 Irastorza, Ramiro Miguel. CONICET.

Teaching, Research & Academic Service

Teaching Experience

- **Full Professor (Tenured)** at Universidad Tecnológica Nacional (La Plata) and former Assistant Professor (Buenos Aires).
- Extensive undergraduate teaching experience at the Universidad Tecnológica Nacional (UTN), National University of La Plata (UNLP), and National University of Rosario (UNR) since 1992.
- Instructor of postgraduate courses focusing on the Physics of Granular Materials and Computational Tools for Scientists.

Research Projects

- **Principal Investigator and Collaborator** on numerous research projects funded by CONICET, the National Agency for the Promotion of Research, Technological Development and Innovation (Agencia I+D+i), and UTN.
- Participant in international research consortia, including projects funded by the European Commission (Horizon 2020).

Institutional Service & Evaluation

- **Scientific Reviewer** for leading international journals in Physics and Materials Science.
- **Grant Reviewer and Evaluator** for national funding agencies (ANPCyT, CONICET) and various National Universities.
- Active involvement in Academic Committees and institutional management roles.

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